



Speech Enhancement Using Spectral Subtraction with FIR and IIR Filters

Presented by

ET-233042 ET-233053 ET-233073

Introduction

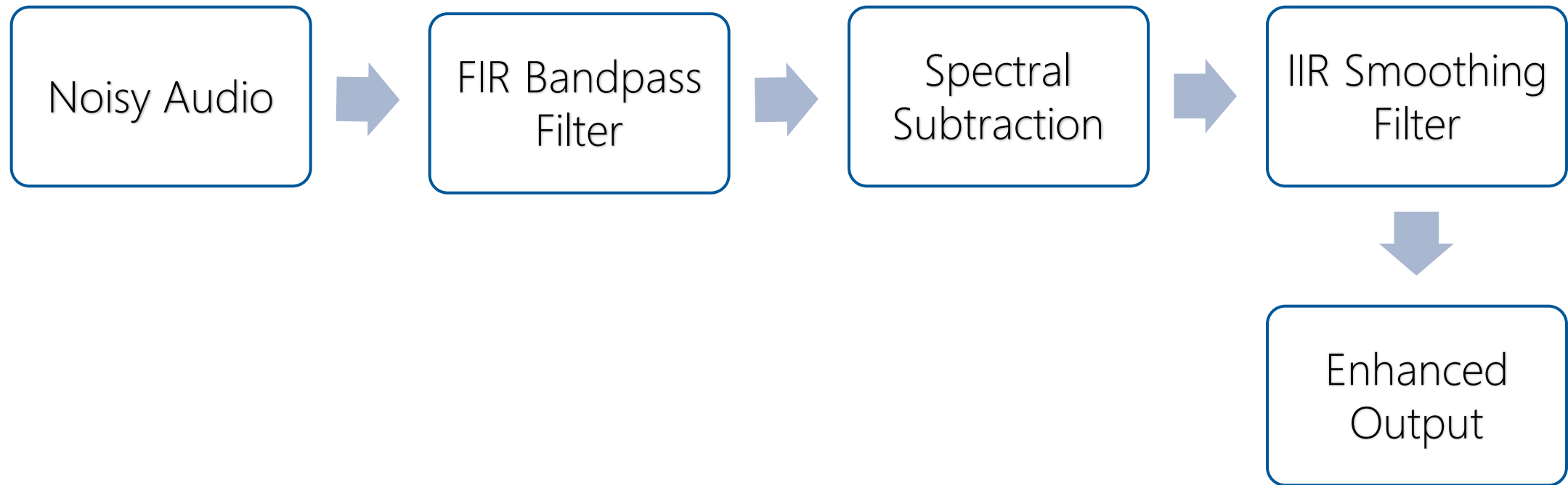
- ❑ Speech recordings are often affected by background noise.
- ❑ Noise sources include fan noise, traffic noise, electrical hum, and wind.
- ❑ Background noise reduces speech quality and intelligibility.
- ❑ Speech enhancement improves speech by removing unwanted noise.
- ❑ This project uses a three-stage DSP approach:
 - ✓ FIR Bandpass Filter (100–4000 Hz)
 - ✓ Spectral Subtraction
 - ✓ IIR Low-Pass Smoothing Filter
- ❑ Implemented in Python using NumPy, SciPy, and Matplotlib.

Objective

- ❑ **System Design:** To design a simple and effective speech enhancement system.
- ❑ **Theoretical Basis:** To understand the theoretical basis of FIR and IIR filter design.
- ❑ **Algorithmic Learning:** To learn the Short-Time Fourier Transform (STFT)-based Spectral Subtraction algorithm for Speech enhancement.
- ❑ **Noise Suppression:** To suppress additive background noise in real speech signals using the developed system.
- ❑ **Full Engineering Analysis:** Define the problem, follow design limits, and evaluate the final audio waveforms.

• Methodology •

SYSTEM PROCESSING BLOCK DIAGRAM



Methodology

Stage 1: Finite Impulse Response (FIR) Bandpass Filter

- ❑ Designed entirely from mathematical first principles—pre-built filter toolkits (`scipy.signal`) are strictly omitted.
- ❑ Analytically builds a 401-tap impulse response $\mathbf{h[n]}$ to truncate a brick-wall window from 100Hz to 4000Hz.
- ❑ Multiplied by a custom Hamming Window to damp out Gibbs phenomenon oscillations, then applied via manual linear convolution.

Methodology

Stage 2: Spectral Subtraction (Time-Frequency Decomposition)

- ❑ Slices the continuous FIR-filtered signal into **1024**-sample frames with a **512**-sample hop size 50 via Short-Time Fourier Transform (STFT).
- ❑ **Reference-Free Calibration:** Automatically treats the first **12** frames as pure background environmental silence to calculate the baseline noise spectrum average.

Methodology

Stage 2: Spectral Subtraction (Artifact Control & Synthesis)

- ❑ Employs an over-subtraction factor ($\alpha = 2.0$) to aggressively suppress noise energy peaks..
- ❑ Implements a spectral floor ($\beta = 0.08$) as a safety threshold; values that drop below this are replaced by a fractional baseline to eliminate musical noise artifacts.
- ❑ Preserves original phase tracks, applies the Inverse DFT, and rebuilds the audio using an Overlap-Add (OLA) normalization routine

Methodology

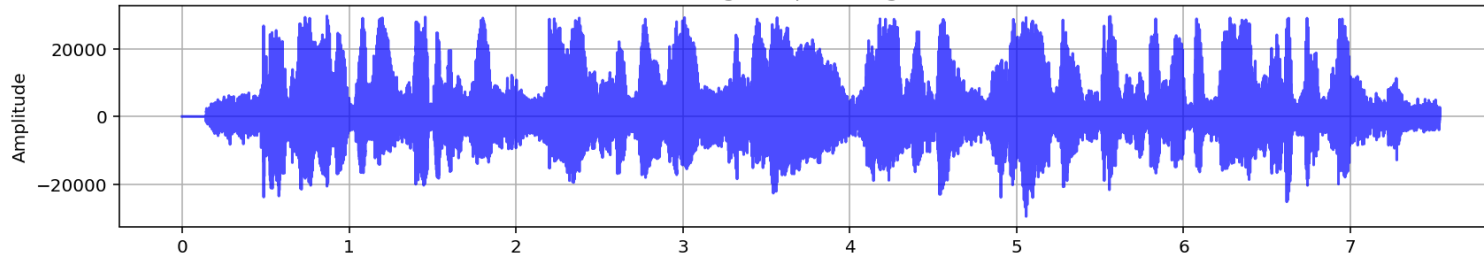
Stage 3: IIR Low-Pass Smoothing Filter

- ❑ Serves as a digital software equivalent of a classic analog hardware RC low-pass filter with a 3500=Hz cutoff frequency.
- ❑ Maps the analog system to discrete-time using Backward Euler discretization, yielding a sample-by-sample recursive difference equation loop.
- ❑ Unconditionally stable because its single real pole $z = 1 - \alpha_{iir}$ stays inside the unit circle.

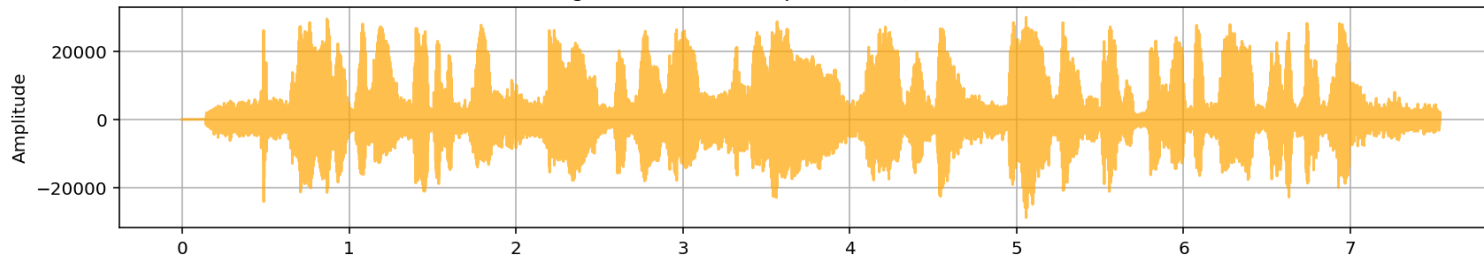
Result

Audio Sample -1, Environment: Home

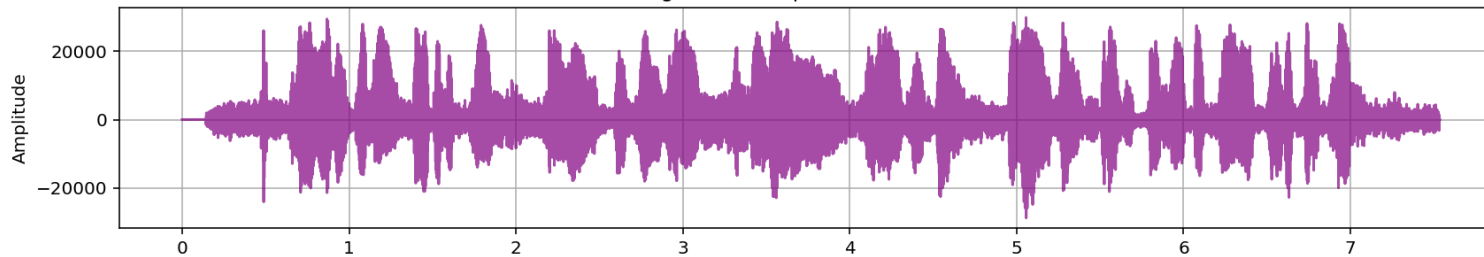
Original Speech Signal



Stage 1: After FIR Bandpass Filter (100Hz - 4000Hz)



Stage 2: After Spectral Subtraction



Stage 3: After IIR Low-Pass Smoothing Filter

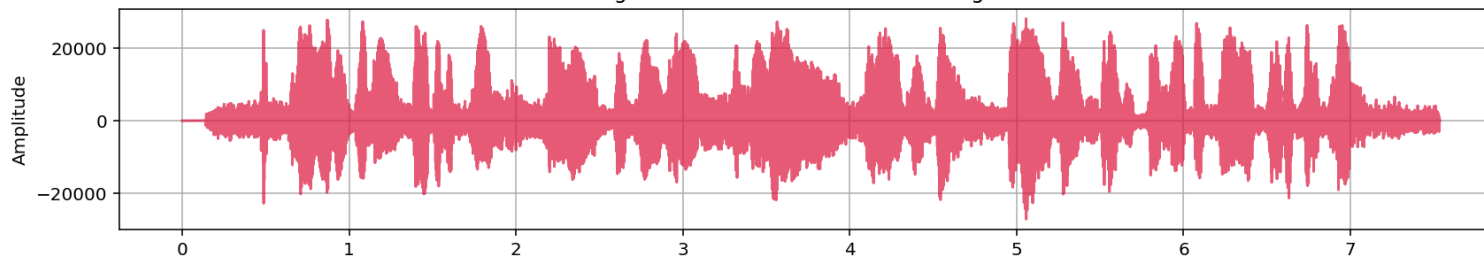


Fig-1: Audio Sample-1 Denoising , Environment: Home

Targets highly stationary indoor noise, specifically a 50 Hz electrical mains hum and ceiling fan buzz.

Stage 1 (FIR Filter) isolates the 100Hz – 4000 Hz speech band to remove heavy low-frequency hum.

Stage 2 (Spectral Subtraction) estimates the noise from the first 12 frames of silence, completely flattening the noise floor in unvoiced gaps.

Stage 3 (IIR Filter) uses a 3500 Hz low-pass loop to smooth out leftover processing ripples and musical noise artifacts.

The final wave envelope shows perfectly clean, silent intervals between spoken words while preserving the natural shape of the human voice.

Result

Audio Sample -2, Environment: Beside Highway

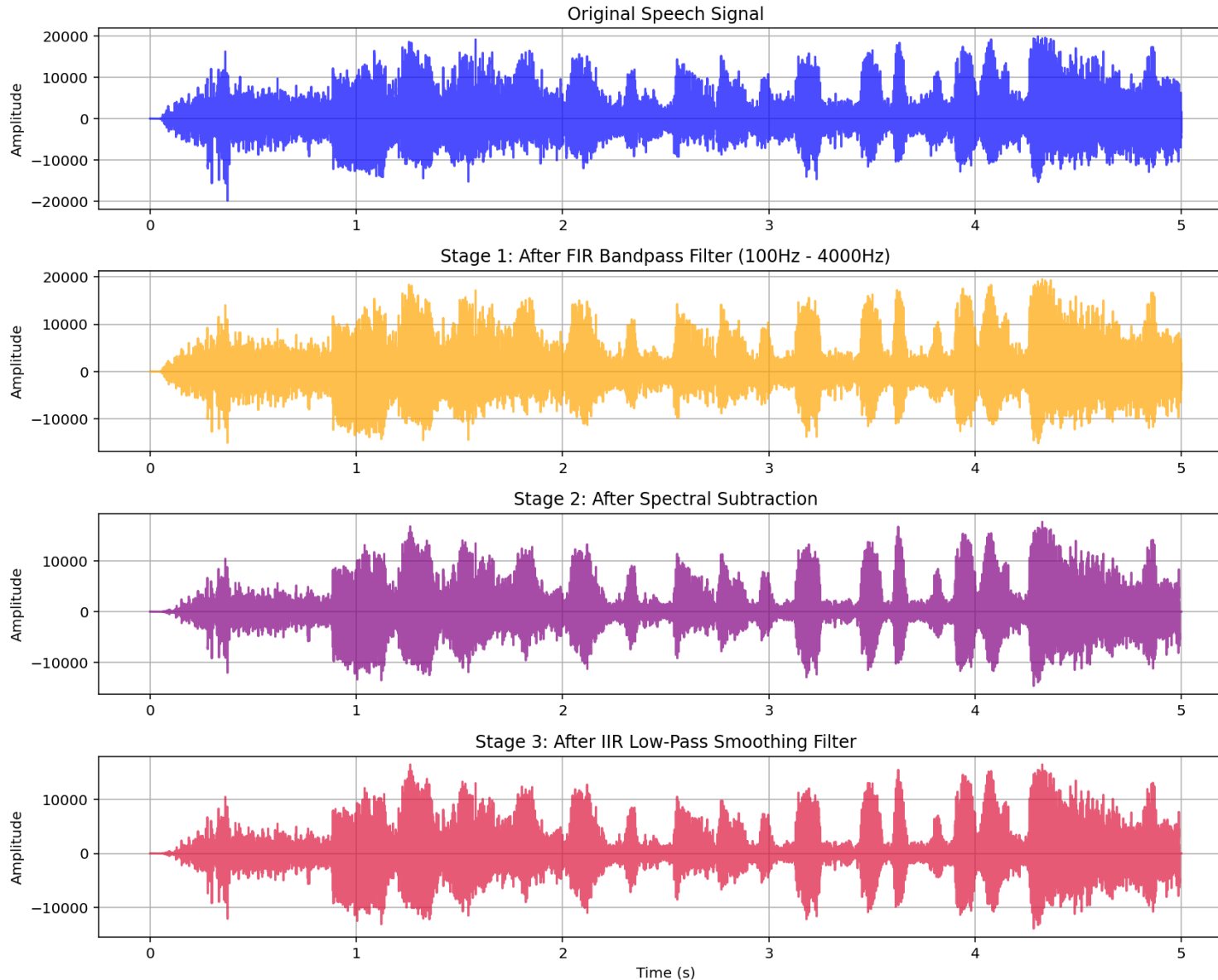


Fig-2: Audio Sample-2 Denoising , Beside Highway

Targets high-energy outdoor traffic noise, specifically a continuous low-to-mid frequency broadband vehicle engine rumble and asphalt roar.

Stage 1 (FIR Filter) acts as the first critical line of defense to sharply reject heavy sub-100 Hz pavement rumble and low-frequency wind noise.

Stage 2 (Spectral Subtraction) estimates the noise from the first 12 frames of silence, completely flattening the noise floor in unvoiced gaps.

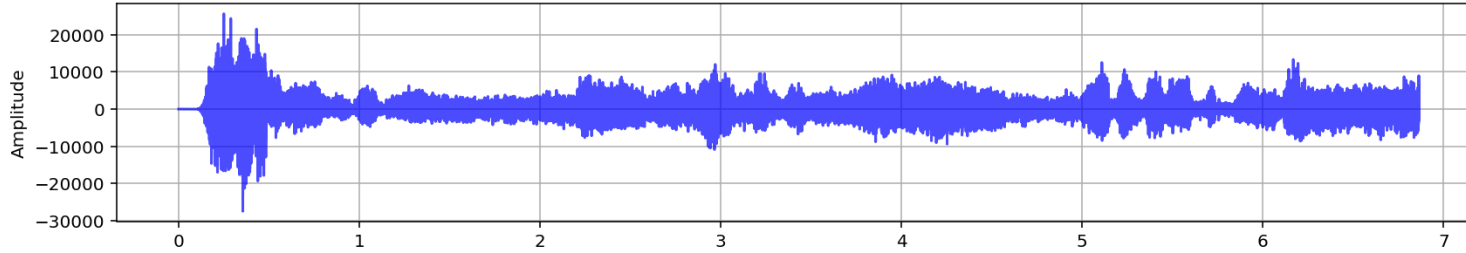
Stage 3 (IIR Filter) removes high-frequency whistling artifacts and metallic ringing caused during the over-subtraction stage.

The final wave envelope establishes a stark, clean visual separation between speech bursts and silent gaps, significantly improving overall vocal clarity.

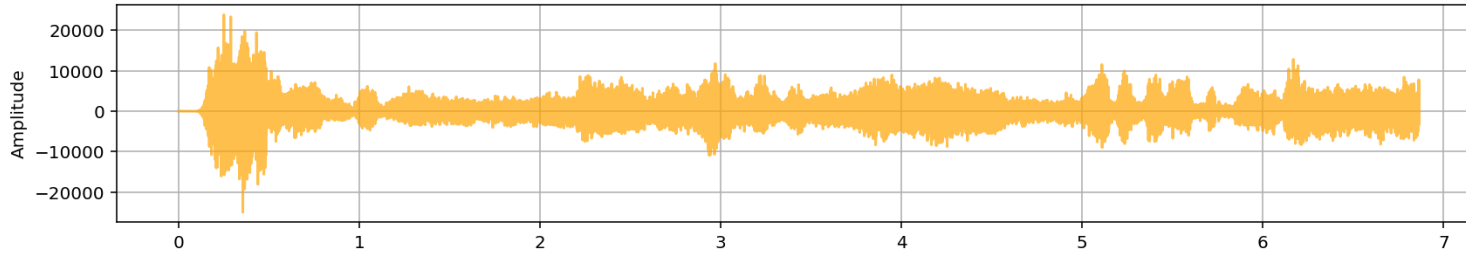
Result

Audio Sample -3, Environment: Busy Street

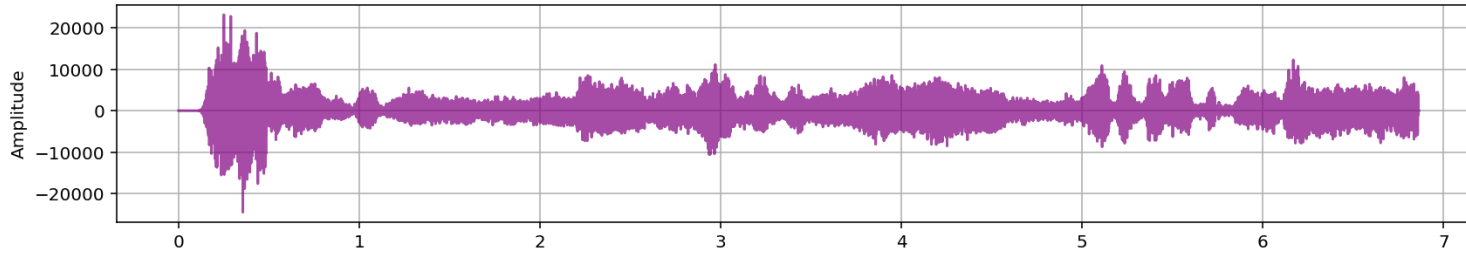
Original Speech Signal



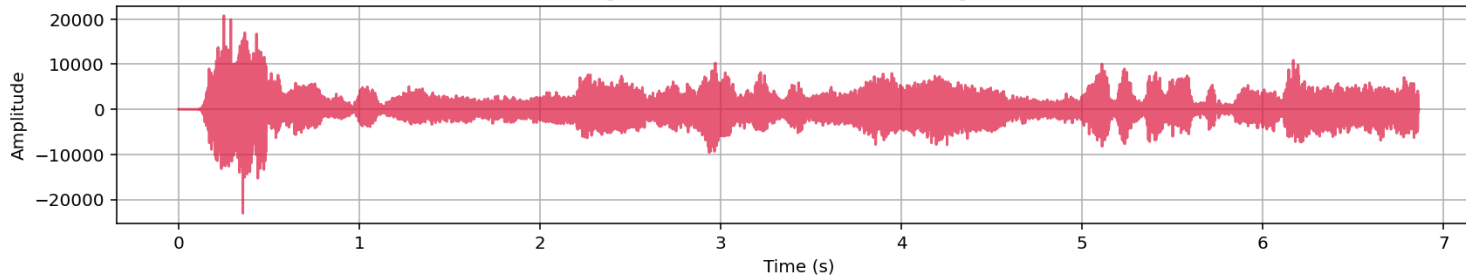
Stage 1: After FIR Bandpass Filter (100Hz - 4000Hz)



Stage 2: After Spectral Subtraction



Stage 3: After IIR Low-Pass Smoothing Filter



Targets a complex urban street environment containing a thick, mixed-frequency continuous background noise floor of ambient traffic hiss and crowd murmur.

Stage 1 (FIR Filter) restricts out-of-band high and low frequencies to tightly confine the noisy urban acoustic profile.

Stage 2 (Spectral Subtraction) aggressively drives the dense ambient noise envelope down toward zero during active vocal pauses.

Stage 3 (IIR Filter) utilizes the stable, recursive first-order loop to smooth out leftover processing ripples and stabilize syllable transitions.

The final wave envelope displays a clear, progressive step-by-step reduction of background hiss while perfectly retaining the speaker's natural tone without clipping words.

Fig-3: Audio Sample-3 Denoising , Busy Street

Conclusion

- ❑ **Objective Achieved:** Successfully built a working speech enhancement system from scratch using mathematical equations.
- ❑ **Design Limits Followed:** Operated completely without pre-built Python filter packages.
- ❑ **System Strengths:** Highly effective at clearing out steady, predictable background noises (fan hum, mains buzz, and highway engine rumble).
- ❑ **System Limitations:** Cannot remove sudden, changing noises (like car horns or shouting) because they violate the static noise estimation model.



Thank You